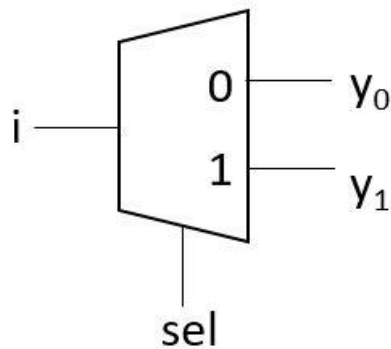


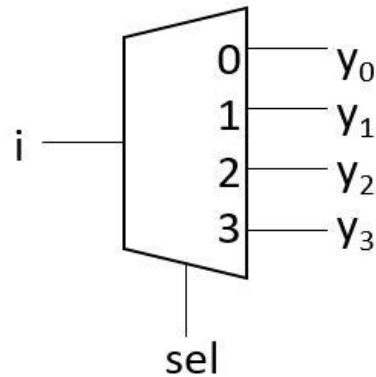
Day 13

1:2 and 1:4 Demultiplexers in Verilog

Circuit Diagram :



1:2 DEMUX



1:4 DEMUX

Verilog Code

A) 1:2 Demux code

```
1 module demux_1x2 (I,sel,y0,y1);
2     input I,sel;
3     output y0,y1;
4     assign {y0,y1}=sel?{1'b0,I}:{I,1'b0};
5 endmodule
6
```

B) 1:4 Demux code

```
1 module demux_1x4 (I,sel,Y);
2     input I;
3     input [1:0]sel;
4     output reg [3:0]Y;
5     always @(*) begin
6         Y=4'b0000;
7         case(sel)
8             2'b00:Y[0]=I;
9             2'b01:Y[1]=I;
10            2'b10:Y[2]=I;
11            2'b11:Y[3]=I;
12        endcase
13    end
14 endmodule
15
```



Testbench Code

A) 1:2 Demux Tb

```
1 module tb_demux_1x2;
2   reg I, sel;
3   wire y0, y1;
4   demux_1x2 dut (I,sel,y0,y1);
5   initial begin
6     $monitor("Time = %0t I = %b sel = %b y0 = %b y1 = %b", $time, I, sel, y0, y1);
7     I = 0;
8     sel = 0; #10;
9     sel = 1; #10;
10    I = 1;
11    sel = 0; #10;
12    sel = 1; #10;
13    $finish;
14  end
15
16 endmodule
```

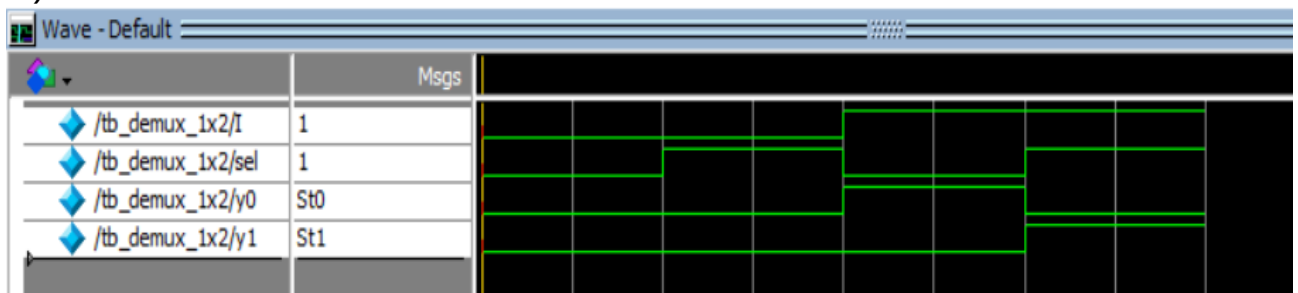
B) 1:4 Demux Tb

```
1 module tb_demux_1x4;
2   reg I;
3   reg [1:0] sel;
4   wire [3:0] Y;
5   demux_1x4 dut(I,sel,Y);
6   initial begin
7     $monitor("Time = %0t I = %b sel = %b Y = %b", $time, I, sel, Y);
8     I = 1;
9     sel = 2'b00; #10;
10    sel = 2'b01; #10;
11    sel = 2'b10; #10;
12    sel = 2'b11; #10;
13    I = 0;
14    sel = 2'b00; #10;
15    sel = 2'b01; #10;
16    sel = 2'b10; #10;
17    sel = 2'b11; #10;
18    $finish;
19  end
20 endmodule
```

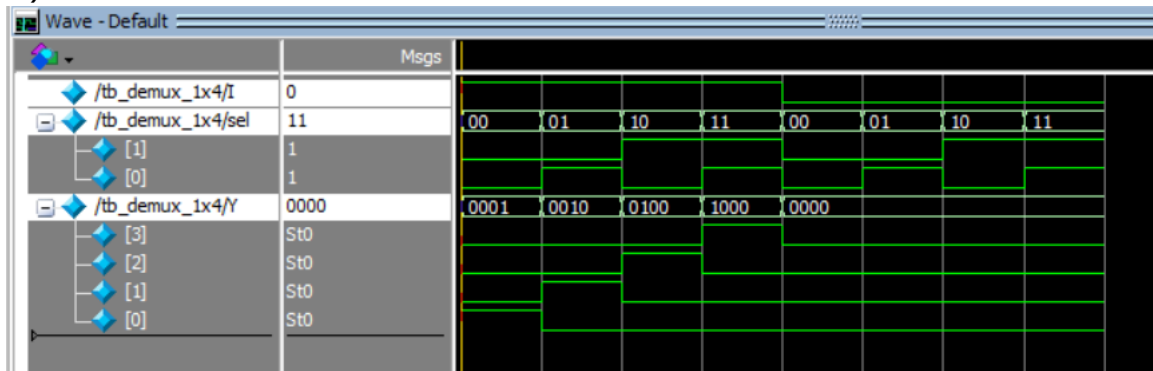


Waveform

A)

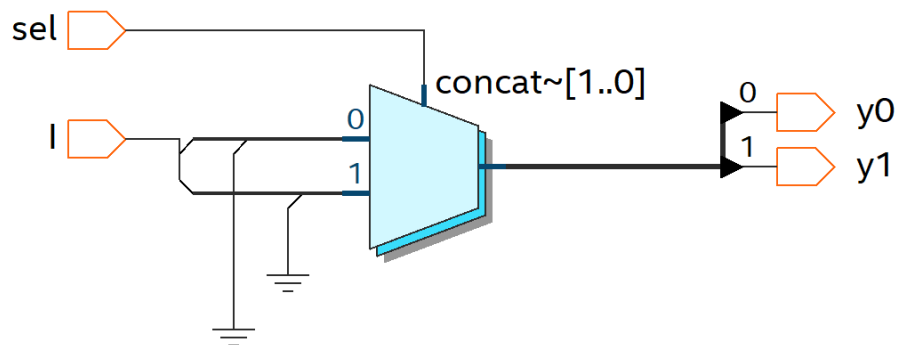


B)

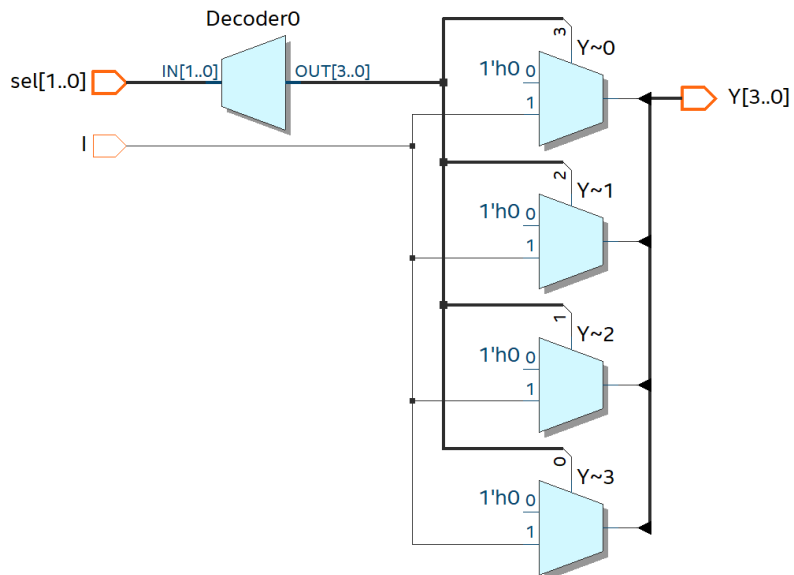


Schematic

A)



B)





EDA Tools Used

- IntelQuartusPrime
- ModelSim

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